

LocalRCA: On-Device Root Cause Analysis for Windows Endpoints Without Ground Truth

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Abstract—Diagnosing why a single Windows machine slowed down or failed is poorly served by fleet observability tools, which assume a network, a backend and consent to export telemetry. We present LocalRCA, a desktop system that records telemetry without sending any of it off the machine, learns that machine’s normal behaviour with an LSTM autoencoder, and explains anomalies with Granger causality constrained by false-discovery-rate control, an effect-size floor and a subsystem topology prior. A personal machine has no labelled incidents, so we evaluate with seven controlled fault injections and a replay over 596 incidents found in real collected history. The system detects and correctly attributes injected CPU, disk and memory faults, and in one case explained a fault it was never told about. On real history it produces a supported causal chain for 106 of 596 incidents (17.8%); 36% are too short to test. The hand-written subsystem prior rejects 22.7% of statistically accepted pairs, including mechanisms that demonstrably exist. We argue that the most useful property of such a tool is saying when it cannot explain an incident.

Index Terms—root cause analysis, anomaly detection, LSTM autoencoder, Granger causality, endpoint telemetry, false discovery rate, explainability

I. INTRODUCTION

Diagnosing why a personal computer became slow, stalled or crashed is typically a manual exercise: an engineer lines up resource graphs against the Windows Event Log and guesses. Fleet-oriented observability platforms automate this, but they assume infrastructure a single machine does not have: a collector network, a time-series backend, and the user’s consent to export telemetry that shows which applications they run.

Targeting a single endpoint imposes three constraints, and most of the design follows from them.

No egress. Telemetry about application usage is sensitive, so the system transmits nothing. The collector contains no network code, which means privacy does not depend on a setting that someone could change.

No ground truth. A personal machine has no labelled incidents. The model cannot be trained to recognise known faults and cannot be evaluated against a labelled test set. This constraint shaped the evaluation more than any other. The only workable response we found was to manufacture ground truth by causing faults on purpose (Section V).

Cold start is unavoidable. “Normal” has to be learned from the machine itself, so the tool can do nothing until it has

seen enough normal behaviour. In our configuration that takes about 21 hours of clean collection.

Our contributions are:

- 1) An end-to-end on-device RCA pipeline combining reconstruction-error anomaly detection with statistically constrained Granger causality (Section IV).
- 2) A fault-injection harness that manufactures ground truth on a machine that has none, and a demonstration that the pipeline can identify a cause it was not told about (Section V).
- 3) A population-scale measurement of how often the causal layer produces an answer at all, replacing claims previously resting on single runs (Section VI).
- 4) An audit showing that the hand-written subsystem prior removes more statistically accepted pairs than any later stage of the pipeline, and that it forbids physically real mechanisms (Section VII).
- 5) A deployment design for a tool that holds sensitive telemetry (per-user encryption of the learned model, signed binaries, and an interface that separates the verdict from the evidence), including where it falls short (Section VIII).

II. RELATED WORK

Anomaly detection in multivariate time series. Surveys of the field [1], [2] separate detectors by whether labels exist and by whether the anomaly is a point, a subsequence or a whole series. Reconstruction-based detectors built on LSTMs [3] are a standard approach when labelled anomalies are unavailable [4], [5]. They train on normal data only and threshold the reconstruction error, so they need no labelled faults, which an endpoint does not have. Later detectors improve accuracy with stochastic recurrent models (OmniAnomaly [6]), adversarially trained autoencoders (USAD [7]), signature matrices fed to convolutional LSTMs (MSCRED [8]), learned sensor graphs with attention (GDN [9]) and transformers (TranAD [10]). MSCRED and GDN also point to the sensors responsible for an anomaly; none of them says why those sensors moved. We keep the plain LSTM autoencoder because its per-metric reconstruction error maps directly onto the metric to be explained and it trains on a commodity CPU. Whether the heavier detectors would pay for themselves on a single endpoint is not tested here.

Log-based anomaly detection. System logs are the other common signal. Xu et al. [11] applied PCA to message-count vectors mined from console logs, He et al. [12] benchmarked six supervised and unsupervised log detectors, and DeepLog [13] learns log-key sequences with an LSTM and flags keys it did not predict. On a personal machine the Windows Event Log is sparse (our host logged 6,600 allow-listed records in 59 days), so LocalRCA uses it only as context around a metric anomaly and not as a detector.

Causal discovery from observational data. Granger causality [14] tests whether one series improves prediction of another beyond the second’s own history. It is correlational under a temporal constraint rather than interventional, and is well known to over-report on correlated series without correction. We apply Benjamini–Hochberg false-discovery-rate control [15] across the full set of ordered pairs, an effect-size floor, and a domain prior over subsystems. Nonlinear extensions replace the linear autoregression with neural networks whose sparse input weights identify Granger causes [16], and PCMCi [17] conditions on each series’ own past and a selected parent set to suppress false links caused by autocorrelation. Both need longer series than an incident window usually provides: Section VI shows that 36% of real incidents are too short even for the linear test.

Root cause analysis in distributed systems. Microservice RCA work typically exploits service call graphs or distributed traces to constrain candidate causes [18], [19], and Soldani and Brogi [20] survey anomaly detection and root cause analysis across service-based cloud applications. A common pattern is to build a dependency graph and walk it. CloudRanger [21] learns a causal graph with the PC algorithm and ranks candidates by a second-order random walk; AutoMAP [22] builds behaviour graphs over several metrics; MicroRCA [23] runs personalised PageRank over an attributed service–host graph; and MicroCause [24] pairs a PCMCi-derived causal graph with a temporal cause-oriented random walk. CIRCA [25] derives a structural graph from the system architecture and treats root-cause identification as recognising which variable was intervened on, while BARO [26] avoids a graph and scores metrics with multivariate Bayesian online change-point detection. A single endpoint has no service topology, so we write an explicit subsystem map by hand, and Section VII shows how much the quality of that map then matters.

Benchmarked causal RCA at fleet scale. Recent work evaluates causal discovery for root cause analysis at a scale this system does not attempt. Ikram et al. [27] discover root causes across thousands of microservice nodes and report top- k recall against ground truth gathered from a real production incident corpus. Pham et al. [28] benchmark six causal-RCA methods (including CausalRCA and BARO) across microservice architectures and find that none dominates uniformly, that performance is highly sensitive to the underlying call graph, and that published recall numbers do not transfer cleanly between systems. Both works assume a call graph or trace corpus that a single endpoint does not have. Our 596-incident survey also lacks comparable ground truth, so Section VI

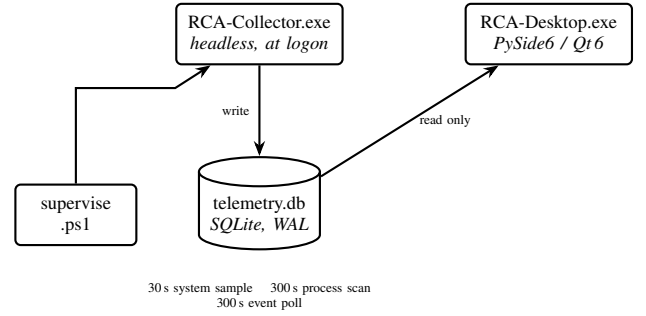


Fig. 1. Process architecture. The collector writes; the desktop application only reads. A PowerShell supervisor restarts the collector if it exits unexpectedly, because creating a Task Scheduler logon task requires elevation that a diagnostic tool should not demand.

reports how often the system gives an answer, not how often that answer is correct.

Positioning. This work differs from the above in that it runs on one machine, has no ground truth to learn from, and is designed to report uncertainty instead of producing a ranking whatever the evidence.

III. SYSTEM ARCHITECTURE

The system is two processes that share a SQLite database (Fig. 1). They are separate because collection has to keep running after the user closes the interface.

The collector samples 29 features every 30 seconds, the top processes by CPU and resident memory every 300 seconds (tightening to 30 seconds under load), and an allow-listed subset of Windows Event Log records. Retention is tiered by sensitivity: numeric readings for 365 days, per-process detail for 30 days, and the foreground application name, the most personal field collected, for 30 days.

At the time of writing the evaluation host had accumulated 124,674 system samples, 1,835,250 process samples and 6,600 Event Log records over 59 days, occupying 227 MB. Coverage of that span is 73.1%: 157 breaks in collection account for 383 hours not recorded. The population survey (Section VI) draws on the most recent 30 days of this history.

IV. METHOD

A. Scaling and windowing

A min–max scaler is fitted on the clean baseline only, $x'_j = (x_j - \min_j) / (\max_j - \min_j)$, and incident windows are transformed with the baseline’s parameters and clipped to $[0, 1]$. Fitting on the incident would normalise away the deviation being sought.

Training windows are built *within contiguous segments only*, never spanning a collection gap. A segment is a maximal run of samples whose spacing is within tolerance of the sampling cadence.

B. Anomaly detection

An LSTM autoencoder compresses a window of T samples over $n = 29$ features to a 32-dimensional latent vector

and reconstructs it. Only the encoder’s final hidden state reaches the bottleneck, so the entire window is represented by 32 numbers before reconstruction; that compression is the detection mechanism. Training minimises

$$\mathcal{L} = \frac{1}{BTn} \sum_{b,t,j} (x_{btj} - \hat{x}_{btj})^2. \quad (1)$$

Error is reduced over time but *not* over features, which is what makes the output attributable to a metric rather than to a window:

$$s_j = \frac{1}{T} \sum_{t=1}^T (x_{tj} - \hat{x}_{tj})^2. \quad (2)$$

Per-metric thresholds are the 99th percentile of validation error, $\tau_j = P_{99}\{s_j(w)\}$, and a metric is anomalous when $s_j > \tau_j$. Per-metric thresholds are necessary because reconstructability varies widely: a near-constant channel reconstructs almost perfectly, while a bursty one does not, and a single global threshold would flag the latter permanently.

C. Constrained causal inference

For each ordered pair of anomalous metrics, both series are differenced until an Augmented Dickey–Fuller test [29] rejects a unit root (at most twice), then tested for Granger causality at lags $1 \dots L$. The lag minimising the p -value is selected. A pair is skipped unless

$$N \geq 3L + 2, \quad (3)$$

which at the default $L = 5$ requires 17 aligned samples. Most of this system’s negative results come from Equation (3).

Surviving p -values are corrected with Benjamini–Hochberg: sorting ascending, the largest rank k satisfying $p_{(k)} \leq \alpha k/m$ is found and ranks $1 \dots k$ accepted, with $\alpha = 0.05$. A p -value alone is not enough, because with a long history even a negligible improvement becomes significant, so we also require an effect-size proxy:

$$\text{effect} = \frac{F}{F + N} \geq 0.10. \quad (4)$$

This quantity is monotone in F , lies in $(0, 1)$, and shrinks as N grows. That is the correction we want, since a large N is what made the p -value small.

Accepted pairs form a directed graph; cycles are removed preferring edges that contradict temporal precedence, and remaining edges are filtered against a subsystem topology prior (Section VII). Candidates are then ranked by a weighted composite plus PageRank [30] on the reversed graph:

$$\text{score}(v) = \min\left(1, 0.70 \sum_k w_k s_k(v) + 0.30 \text{PR}(v)\right), \quad (5)$$

with weights 0.40 causal outflow, 0.30 temporal priority, 0.20 low inflow, 0.05 severity and 0.05 event correlation.

A structural caveat. With zero surviving edges, every node has outflow 0 and inflow 1, so 60% of the weight becomes constant and the ranking collapses onto severity and timing while still emitting a score. The system therefore reports the distinction explicitly rather than presenting the ranking as causal.

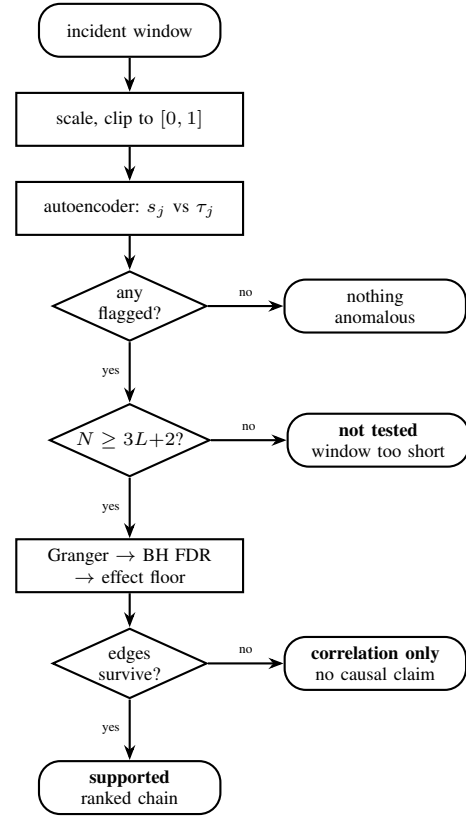


Fig. 2. Inference pipeline. Four terminal states are reported distinctly. Collapsing “not tested” into “nothing survived” would conflate the absence of a result with a negative result, which earlier revisions of this system did.

TABLE I
CONTROLLED FAULT INJECTIONS, 30-MINUTE WINDOWS UNLESS NOTED

Fault	Samples	Flagged	Edges	Outcome
CPU, 7 min	14	6/29	—	not tested (below floor)
CPU	60	6/29	6	explained correctly
Disk	60	4/29	0	detected, unexplained
Memory	60	2/29	0	wrong process named
Memory (refixed)	60	3/29	0	detected, attributed
Memory (clean)	60	0/29	—	not measurable on host
Idle	60	1/29	—	3.4% false positives

V. EVALUATION BY FAULT INJECTION

Because a personal machine provides no labelled incidents, we manufacture ground truth: a harness causes a specific, known disturbance, waits for samples to land at the real cadence, runs the production pipeline over the injection window, and scores the result.

A. A correct explanation of an unknown cause

Half the CPU cores were burned for 30 minutes with no information given to the pipeline. The ranking named `cpu_pct` first at 1.000, and the surviving edges point *away* from CPU. That is what a real root cause should look like, since nothing upstream drives it:

`cpu_pct_max_core → disk_busy_pct`

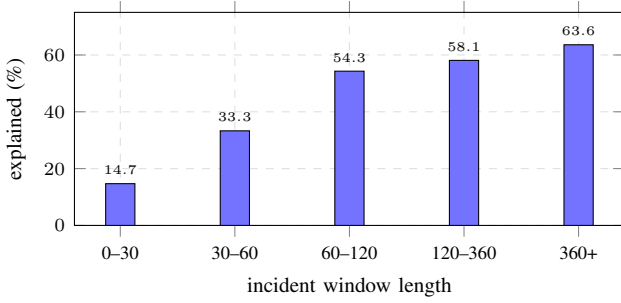


Fig. 3. Causal yield against window length, over 381 analysable incidents ($L = 5$, 30-day survey). The rate rises steadily with window length, which is the short-window effect first seen in a pair of injections, now at scale.

(lag 2, 0.756).

B. Short windows starve the causal layer

The same fault at 7 and at 30 minutes differed only in duration. At 14 samples, against the floor of 17 from (3), *no pair was compared at all*, and the system reported exactly that. At 60 samples, ten pairs survived correction and six survived the topology filter. The causal layer worked; it was short of samples, so the fix is a wider window rather than a redesign. Section VI confirms this at scale.

C. A defect that injection exposed

The memory injection detected the anomaly and then attributed it to four unrelated processes. Attribution ranked candidates by mean CPU, while resident memory was selected but never sorted on, so a process that allocates memory and then sleeps could never be named. A memory-bound cause was unreportable *by construction* in every incident the system had ever analysed. The CPU and disk faults are both CPU-heavy, so they happened to be attributed correctly, and that hid the defect.

D. A measurement this host cannot provide

A later memory injection on a deliberately cleared machine flagged nothing. The detector was correct: the host’s median memory utilisation across 30,199 samples is 95.2%, and the injection window sat at 87.4%, the 9th percentile of the host’s own history. For safety the injection is limited to half of free memory, and on a host that is usually full that cannot push usage above the level the machine already reaches on its own. The fault is undetectable *by construction* here, and the harness now reports this as inconclusive rather than as a detector failure.

VI. POPULATION-SCALE EVALUATION

Fault injection establishes correctness on a handful of cases but yields single observations. To measure *frequency*, the pipeline was replayed over every incident found in real collected history, read-only and with no injection.

Of 596 incidents discovered, 215 (36%) fall below the sample floor of (3) and cannot be tested at any setting. Of the 381 that can be, 106 produce a supported causal

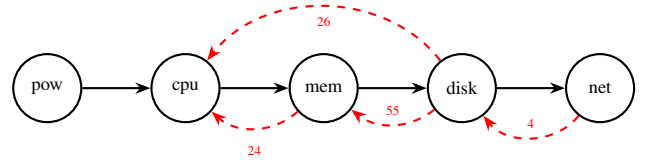


Fig. 4. The subsystem prior is a strict total order (solid), so *network* is a sink and nothing can be reported as caused by network activity. Dashed edges are directions the statistics accepted and the map rejected, labelled with occurrence counts (30-day, 596-incident survey). Two of them describe real mechanisms: the file cache consuming memory under write pressure, and interrupt handling costing CPU during disk activity; a download writing to disk (*network*→*disk*) remains a smaller but real third case.

chain. The end-to-end figure a user experiences is therefore $106/596 = 17.8\%$: roughly one incident in six receives a causal explanation.

We report the whole funnel. Counting only *analysable* incidents would give 27.8%, which looks better but describes fewer situations.

The floor is a parameter artefact. Short incidents are widened before analysis to exactly the model’s window size, 15 samples, while (3) at $L = 5$ requires 17. The two constants were chosen independently in the same pipeline. Re-running the survey at $L \in \{3, 4, 5\}$ over an identical population (598 incidents discovered over 30 days of real collected history) confirms the cliff disappears at $L \leq 3$: 0 incidents fall below the floor, versus 215 at $L = 5$. Explanations rise only modestly, from 106 to 110, while the tested population nearly triples (381 to 598), and median surviving edges holds at 2 throughout. What improves is the reporting: more incidents read “tested, nothing found” instead of “not tested”, but few gain an explanation.

VII. AUDITING THE SUBSYSTEM PRIOR

Since no service call graph exists on an endpoint, the system encodes domain knowledge as an explicit map of permissible subsystem influence. Auditing it across all analysable incidents shows it is the largest single filter in the pipeline after the statistical gates: of 538 accepted pairs, 337 (62.6%) survive, 122 (22.7%) are rejected by the map and 79 (14.7%) by cycle-breaking.

Every rejected transition has *zero* survivors, because the map forbids whole classes rather than individual edges (Fig. 4). Judged individually, *disk*→*memory* is almost certainly real (the Windows standby file cache grows under write pressure) and is the strongest single rejected relationship measured in this run, at 0.654 (*disk_write_bps*→*swap_used_delta*, lag 2). *disk*→*cpu* reflects interrupt and DPC handling, and *network*→*disk* is the download-writes-to-disk case argued in the prior revision of this analysis, present but smaller in this run (4 occurrences). Conversely, the smallest rejected classes (*network*→*process*, *disk*→*process*, at 1–2 occurrences each) are plausibly spurious rather than real mechanisms being wrongly forbidden.

The problem with the prior is that it only runs one way. It models resource pressure flowing downhill and cannot express the feedback that makes I/O expensive. Adding the reverse edges would make the graph cyclic, and cycle-breaking already removes 14.7% of pairs, so the two mechanisms would start to compete. We leave this as an open design question instead of patching it.

VIII. DEPLOYMENT ENGINEERING

A research prototype can leave its data in plaintext and show its results as raw tables. A tool that someone else installs, and that records when and how they use their machine, should not. This section describes what the released application (v1.5.1) does about both problems and where it falls short.

A. Protecting the model at rest

The trained artefact holds the scaler’s per-metric bounds and the per-metric thresholds, which together form a compact statistical profile of how the machine is used. It is wrapped with the Windows Data Protection API (DPAPI), which binds the ciphertext to the user’s logon credentials, so a copy taken to another account or machine cannot be read. The write is atomic, via a temporary file and a rename, so an interrupted save cannot leave a truncated model. Artefacts from earlier versions carry no header, still load, and are encrypted the next time the model is trained. If the user’s credentials are reset and the key is lost, the application says the model must be retrained rather than surfacing a raw Windows error. The database and logs, which DPAPI does not wrap, are covered by a second, best-effort layer: at start-up the collector asks the Encrypting File System (EFS) to encrypt its data folder.

Three findings qualify this. First, EFS does not exist on Windows Home editions. The evaluation host runs one, the request fails on every start, and its 227 MB database remains in plaintext, protected only by file-system permissions. The failure is logged rather than silent, but the protection a user receives depends on which edition of Windows they bought. Second, re-encryption happens only on retraining, so the model on the evaluation host, which was trained before encryption was added, is still in plaintext. Third, independent review of the encryption change found that training wrote an intermediate checkpoint in plaintext and deleted it only on success; an interrupted run left unencrypted weights on disk. The checkpoint is now removed in a `finally` block, and any leftover from an earlier crash is purged when training starts.

B. Signed binaries

Both executables are Authenticode-signed with an RFC 3161 timestamp, and each release publishes a SHA-256 checksum. The certificate is self-signed: it shows that a binary is unchanged since it was built, not who built it, so Windows SmartScreen still warns on first run. Removing the warning requires a certificate from an authority Windows trusts, which this project does not have.

C. Separating the verdict from the evidence

A review of the running application found that the first screen presented a table of every captured channel with its raw latest value, and that training readiness was expressed four different ways (clean samples, longest run, current run, time remaining) that a non-specialist had to reconcile alone. Hiding these would contradict the project’s premise, since a user cannot consent meaningfully to collection they cannot inspect.

The released interface therefore presents the same computation in two ways, selected by a single global switch that persists across sessions and is off by default. In the default view each tab leads with one sentence whose colour encodes state: the data tab reports *Healthy* only when coverage is at least 90% with at most three breaks, and *Partial data* from 50%; the training tab states whether a model can be trained and, if not, how long remains. The inference tab keeps the incident picker and opens on the plain-language report rather than the score table. The advanced view restores the raw counts, the channel table, the database path, the training hyperparameters, the Granger lag, and the score and outflow columns. Nothing is computed differently in either view; the switch changes presentation only, and the headline sentence is derived from the same coverage figures the advanced view prints. The evaluation host itself shows *Partial data* at 73% coverage, which is correct.

This design was checked by one expert review and by driving the live application. It has not been measured with users, and whether the default view helps a non-specialist reach a correct conclusion is unestablished.

IX. DISCUSSION AND THREATS TO VALIDITY

All measurements come from one machine. Nothing here shows that the results generalise. One measurement, memory-fault detection, cannot be made on this host at all, so testing on a second machine is required, not optional.

Correctness is not established at population scale. The survey’s 596 incidents have no known cause, so the 17.8% explanation rate says how often the causal layer gives an answer, not whether the answer is right. It is an upper bound on usefulness rather than a measure of accuracy.

No fault has been tested whose cause is not also the loudest metric. Both explaining runs put the injected fault atop a severity ranking, so a correct causal answer and a correct severity answer remain indistinguishable.

Granger causality is not causation. It is correlation under a temporal constraint. The FDR correction, effect floor and subsystem prior exist to suppress the resulting over-reporting, and Section VII shows the prior itself is imperfect.

A methodological observation. Nine defects were found in code carrying a passing test suite of 130 tests. In nearly every case the test checked something next to what mattered: a database value instead of the bytes on disk, one file instead of the full storage footprint, or a function instead of the code path that calls it. Two fixes were worse than the defects they repaired. These were caught by adversarial review rather than

additional testing, which we believe generalises beyond this system. The pattern recurred once encryption was added: two reviewers independently found the plaintext training checkpoint of Section VIII in code whose tests all passed, because the tests exercised only the successful path. The suite has since grown to 195 tests.

X. CONCLUSION

We presented an on-device root cause analysis system for Windows endpoints operating without ground truth, network access or fleet context. Under controlled injection it detects and attributes CPU, disk and memory faults, and has produced a correct causal explanation of a fault it was never told about. Across real history it explains roughly one incident in six, declines to explain the rest, and distinguishes “not tested” from “tested and nothing survived”.

We argue this last property matters more than the explanation rate. A tool reporting “no causal chain was supported” when the data cannot support one is more useful than a confident ranking that changes with the analysis window. In this system, getting it to say so honestly took more engineering than computing an answer did.

Future work: repeating the evaluation on second hardware, testing a fault whose cause is not the loudest metric, resolving whether the subsystem prior should admit the feedback directions it currently forbids, encrypting the database independently of EFS so that Home-edition users receive the same protection, and measuring with users whether the default view leads them to correct conclusions.

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